# Alhassan Tameem Amer

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## PROFILE

Software Engineering graduate from Cairo University's Faculty of Computing and Artificial Intelligence, with a broad foundation across backend systems, algorithms, and full-stack development built through both coursework and hands-on projects. I enjoy taking a system from architecture to working implementation — most recently leading backend development on a full-stack platform where I designed the API structure, implemented security, and coordinated a team through delivery.

My technical range spans Java, Python, C++, and SQL, with practical experience in Spring Boot, PostgreSQL, and REST API design, alongside earlier work in Java EE, mobile development, and classic data structures and algorithms. I'm equally comfortable working independently on a focused technical problem or coordinating with a team across the full stack.

I'm looking for a role where I can keep growing as an engineer while contributing to systems that need to actually hold up under real use.

**Part time Exam Invigilator** **April 2025 – May 2026**
British Council

## EDUCATION

**Bachelors, Software Engineering** **October 2022 – June 2026**
Faculty of Computer Science and Artificial Intelligence, Cairo University

**Digital Egypt Pioneers Initiative – Mobile App Development Track** **April 2024 – October 2024**

**Al Ahli Bank of Kuwait – IT Internship** **July 2025**
System Administration for 1 week and Development Team (C# & .NET) for 3 weeks

## PROJECTS

**Eventra — Smart Event Management Platform** **2025–2026**
Graduation Project (Backend Development)

- Led backend development of a full-stack event management platform for the MENA region, architecting a modular Spring Boot REST API across 6 independent domains: Authentication, Event Management, Booking & Payment, Notifications, Gamification, and AI Analytics.
- Designed and implemented JWT-based authentication with role-based access control (Attendee, Organizer, Admin), including refresh token rotation, email verification, and Redis-backed rate limiting to prevent brute-force attacks.
- Built a complete event lifecycle system covering creation, multi-stage admin approval, discovery with paginated search/filtering, and a booking engine handling ticket holds, payment processing, and partial refunds with automatic capacity management.
- Provided cross-functional support across the frontend (React/TypeScript) and AI microservice (Python/FastAPI) teams, assisting with API integration, debugging, and ensuring consistency between backend contracts and client-side implementation.
- Conducted load testing using k6 to validate API performance under concurrent traffic and verify booking logic correctly prevents ticket overselling during simultaneous transactions.
- Coordinated a 5-person team using a Git branching strategy (feature branches → dev → main) with PostgreSQL schema migrations managed through Flyway across all modules.
- Used tools: Java 21, Spring Boot, Spring Security, PostgreSQL, Redis, Flyway, Maven, k6, Git/GitHub, Postman, React, FastAPI.
- Used skills: System Architecture, REST API Design, Database Design, Team Leadership, Cross-functional Collaboration, Performance Testing, Security Implementation.

**Loan Installment & Payment Scheduler** **July 2025**
Financial Calculation Project

- Developed a C# console application to calculate loan installments and generate a detailed payment schedule for monthly and quarterly repayment plans.
- Implemented the amortization formula to compute accurate installment amounts, breaking down each period into interest and principal, and updating the remaining balance accordingly.
- Used conditional logic and user input validation to ensure correct loan amounts and payment durations, providing flexibility in choosing between monthly and quarterly plans.
- Output includes a clear, tabulated payment schedule showing all financial breakdowns until the loan is paid off.
- Used tools: C#, .NET Console, Math functions.
- Used skills: Problem Solving, Financial Math, Console UI, Input Validation.

**Excel Transaction Formatter** **July 2025**
*File Processing & Automation Project*

- Developed a C# console application to automate the transformation of structured Excel data into a formatted .txt file for downstream processing, supporting financial reporting and reconciliation.
- Used the ClosedXML library to read .xlsx files and dynamically parse, format, and output data into three structured sections: header, transaction lines, and summary line based on user-specified credit or debit flag.
- Implemented string manipulation and formatting logic to generate transaction records in a strict padded format, including card number, currency, transaction codes, and amounts with validation for totals.
- Handled user input, exception-free reading of Excel data, numeric conversions, and file writing using the System.IO and System.Text libraries.

- Used tools: C#, ClosedXML, .NET Console, Excel.
- Used skills: File I/O, Data Parsing, Problem Solving, Automation, Console UI Design.

**MiniSocial App** **April 2025 – May 2025**
*Java EE Development Project*
- Developed a lightweight social networking web application with features for user registration, profile management, friend requests, post creation, and group management using a modular Java EE architecture.
- Implemented business logic with EJBs, data persistence with JPA and JDBC, transaction management with JTA, RESTful APIs with JAX-RS, role-based security with JAAS, and asynchronous notifications using JMS.
- Used tools and technologies: Java EE, JSP, Servlets, JDBC, EJBs, JPA, JTA, JAX-RS, JAAS, JMS.
- Used Skills: Problem Solving, Communication, Teamwork, Web Development, Project Management.

**Reach Fast Gas Calculator App** **September 2024 - October 2024**
*Mobile Development Project*
- Developed a mobile application with a team of developers to calculate gas costs for trips between multiple destinations and provide route-finding capabilities. Features included gas cost estimation based on distance, vehicle specifications, and fuel prices.
- Utilized location tracking (GeoLocator/AirLocation) for real-time positioning and persistent storage (Hive/SharedPreferences) for saving user data. Implemented geocoding for converting place names to coordinates and integrated user-friendly interfaces with RecyclerView and Toast for feedback.
- Includes a Native Android and a Flutter application.
- Used tools and Libraries**:** Java, Flutter, GeoLocator, Hive, SharedPreferences, Gson, AirLocation, Geocoding API.
- Used skills: Problem Solving, Communication and Teamwork.

**Dijikstra's Shortest Path Algorithm Implementation** **May 2024**
*OOP-Based project*
- Developed a directed weighted graph application implementing Dijkstra's shortest path algorithm using a min-heap for efficient node label and cost management.
- Created and utilized data structures for graph representation, adjacency matrices, and heap operations to calculate and output shortest paths from a specified starting vertex
- Used tools: C++, OOP, Data Structures and Algorithms.

**Hospital Emergency Room Simulation** **April 2024 – May 2024**
*OOP-Based Project*
- Developed a priority queue system to manage patients based on the severity of their conditions, implementing functionalities to add, process, update, display, and retrieve patients. Ensured efficient queue management with priority handling and real-time updates.
- Used tools: C++, OOP, Data Structures and Algorithms.

## SKILLS

Technical: Object Oriented Programming, Python, C++, C#, Java, SQL, Microsoft Office, Data Structures & Algorithms, Docker, .NET, Spring Boot, PostgreSQL, Redis, Git/GitHub.
Soft Skills: Problem Solving, Presentation, Leadership, Project Management, Communication, Critical Thinking, Entrepreneurship, Teamwork, Attention to Detail, Time Management.
Languages: Arabic (Native), English (Fluent)